

September 21, 2026

Dear Hiring Team,

I am writing to apply for the Software Engineer - Data Platform position at Samsara. With a Master of Science in Computer Science from the University of Calgary and hands-on experience building C++ and Python systems for large-scale geospatial data processing, storage, retrieval, and validation, I am excited by the opportunity to contribute to infrastructure that supports Samsara's analytics, AI, product, and operational systems.

What makes this role especially interesting to me is that it is not a dashboard or reporting role. It is about the infrastructure layer that makes large volumes of data usable, reliable, discoverable, and scalable for other teams. That maps closely to my graduate research, where I designed and implemented a DGGS-based framework for managing spatiotemporal Earth observation data. I built C++ and Python pipelines for preprocessing, encoding, storing, and retrieving satellite imagery; designed a tensor-based storage model for direct access to arbitrary regions at multiple resolutions; and worked on temporal retrieval methods for missing timestamps. This required careful thinking about data layout, retrieval performance, metadata, scalability, validation, and the tradeoffs between storage efficiency and access speed.

My research also gave me practical experience improving performance in data-intensive systems. I reduced one encoding workflow from 233 seconds to 26 seconds using multithreaded C++, and reduced storage by 38% on a scaled Canada-wide dataset with 1,112 GeoTIFF tiles and more than 16,000 DGGS tensors. While the domain was Earth observation rather than IoT operations, the engineering shape is similar: moving large data through reliable pipelines, making it accessible for downstream users, and building foundations that allow analytics and machine learning work to happen safely.

I also bring backend and platform-adjacent engineering experience. At Asr Gooyesh Pardaz, I developed Django REST APIs backed by PostgreSQL, built authentication workflows, and implemented payment-service logic. In my current agentic automation project, I am integrating Docker-based services, Google Sheets, Google Drive, Telegram API, LLM workflows, and human review into a system that searches, evaluates, and tracks applications. Across these projects, I have relied on Git, Linux, documentation, debugging, and iterative engineering practices to build maintainable systems.

I recognize that Samsara's ideal candidate brings deeper production experience with Spark, Databricks, Delta Lake on S3, AWS infrastructure, and large-scale operational data platforms. I am still growing in those specific technologies. At the same time, my strongest work has been exactly in the direction of data infrastructure: designing storage and retrieval systems, building processing pipelines, optimizing performance, and supporting data workflows used by other technical users. I would be highly motivated to ramp up quickly on Samsara's stack and contribute from my Python, C++, data-systems, and research-software foundation.

Samsara's mission also resonates with me. I like engineering work where data infrastructure is connected to real-world systems and operational decisions, not just abstract metrics. Thank you for considering my application. I would welcome the opportunity to discuss how my data-platform, geospatial systems, and software engineering background can contribute to Samsara's Data Platform team.

Sincerely,

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